

constraint published product

INPUTS

GRACE / GRACE-FO

JPL RL06 mascons, 3°
19 in-basin, monthly

ERA5-Land

0.1°, daily → monthly
P · ET · runoff · soil water

Covariates

terrain · soil · water table
irrigated cropland

TARGET AND FEATURES

Target decomposition

TWSA = level + trend · t + anomaly
level and trend taken from GRACE: never fitted

Feature construction

storage memory · neighbourhood context
climatology · seasonality

MODEL COMPARISON

Five candidates under one protocol

RandomForest · XGBoost · LightGBM · XGBRF · MLP
Optuna on mascon-grouped CV with fold-level pruning; the MLP is scored at a fixed configuration from a released 21-point sweep, not searched each run ranked on the same folds; the margin is judged against each model's spread

DOWNSCALE AND CONSTRAIN

Spatial downscaling

the selected model is fitted across mascons
and applied at 0.1°

Mass conservation

minimum-norm correction; every mascon mean
reproduced exactly: imposed, not learned

PRODUCTS

0.1° MONTHLY TWSA

2000–2025 · five per-pixel terms
 $\sigma_{\text{GRACE}}, \sigma_{\text{transfer}}, \sigma_{\text{within}}, \sigma_{\text{gap}} \rightarrow \sigma_{\text{total}}$

0.1° DAILY TWSA

water-balance shape; monthly mean preserved
exactly: disaggregated, not downscaled

EVALUATION

Leave-one-mascon-out

19 mascons, each with
its neighbour buffer

Month holdouts

random (optimistic)
blocked · forward

CGWB wells

656 wells, quarterly; per well,
per mascon, basin mean

TreeSHAP

by feature family;
the anomaly only